

LAB 1 – USE-CASE FLOW SPECIFICATION

Local Sports Tournament & Bracket Manager

Use Case ID	UC-001
Use Case Name	Generate Tournament Bracket
Primary Actor	Tournament Director
Goal	Generate a valid seeded single- or double-elimination bracket from registered teams and their rankings.

Preconditions

- The Tournament Director is authenticated and has tournament-management permissions.
- A tournament has been created and its format (single or double elimination) is selected.
- Required teams are registered in the tournament.
- Each registered team has a valid initial ranking or seed.

Postconditions

- A valid tournament bracket is generated according to the selected format.
- Every registered team is assigned to a valid bracket position and match pairing.
- The generated bracket is stored by the system and is available for viewing.

Main Success Scenario

1. Tournament Director selects the tournament and chooses Generate Tournament Bracket.
2. System displays the selected tournament format and retrieves the registered teams and their rankings.
3. Tournament Director confirms the tournament format and bracket generation request.
4. System validates that all required teams and ranking/seed information are present and valid.
5. System orders teams according to the initial rankings and assigns valid seeds.
6. System generates the bracket using the selected single- or double-elimination format.
7. System creates valid match pairings and assigns each team to a bracket position.
8. System stores the generated bracket and its match structure.
9. System displays the completed bracket to the Tournament Director.
10. The Tournament Director can publish the bracket, and Team Captains can view the published bracket.

Alternate Flow – Invalid or Missing Seed Information

- A1. At Step 4, the system detects missing, duplicate, or invalid ranking/seed information.
- A2. System stops bracket generation and displays the specific validation error.
- A3. Tournament Director corrects the team ranking/seed information.
- A4. Tournament Director submits the generation request again.
- A5. System revalidates the information and continues from Step 5 of the Main Success Scenario.